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Exercise 9: Additional Practice Sheet

1. **Purity with different cluster structures.**

   Given are 12 instances and three classes:

   $$A = \{a_1, a_2, a_3, a_4\}, \qquad B = \{b_1, b_2, b_3, b_4\}, \qquad C = \{c_1, c_2, c_3, c_4\}.$$

   A clustering algorithm returns the following two imperfect clusterings.

   Z_A with $K = 3$:

   - $U_1 = \{a_1, a_2, b_1\}$
   - $U_2 = \{a_3, b_2, b_3, b_4\}$
   - $U_3 = \{a_4, c_1, c_2, c_3, c_4\}$

   Z_B with $K = 5$:

   - $V_1 = \{a_1, a_2\}$
   - $V_2 = \{a_3, a_4, b_1\}$
   - $V_3 = \{b_2, b_3\}$
   - $V_4 = \{b_4, c_1, c_2\}$
   - $V_5 = \{c_3, c_4\}$

   Calculate the purity of Z_A and Z_B. Which clustering gets the higher purity?

2. **Rand Index and Jaccard Coefficient without pre-filled matrices.**

   Given are 7 instances $\{r_1, \ldots, r_7\}$ and three classes:

   $$A = \{r_1, r_2, r_3\}, \qquad B = \{r_4, r_5\}, \qquad C = \{r_6, r_7\}.$$

   The clustering result is:

   $$H_1 = \{r_1, r_2, r_4\}, \qquad H_2 = \{r_3, r_6\}, \qquad H_3 = \{r_5, r_7\}.$$

   The class similarity matrix and cluster similarity matrix are given below.

   The class similarity matrix is defined by

   $$\text{ClassSim}(r_i, r_j) = \begin{cases} 1, & \text{if } r_i \text{ and } r_j \text{ belong to the same class,} \\ 0, & \text{otherwise.} \end{cases}$$

	r_1	r_2	r_3	r_4	r_5	r_6	r_7
r_1	–	1	1	0	0	0	0
r_2	1	–	1	0	0	0	0
r_3	1	1	–	0	0	0	0
r_4	0	0	0	–	1	0	0
r_5	0	0	0	1	–	0	0
r_6	0	0	0	0	0	–	1
r_7	0	0	0	0	0	1	–

The cluster similarity matrix is defined by

$$\text{ClusterSim}(r_i, r_j) = \begin{cases} 1, & \text{if } r_i \text{ and } r_j \text{ belong to the same cluster,} \\ 0, & \text{otherwise.} \end{cases}$$

	r_1	r_2	r_3	r_4	r_5	r_6	r_7
r_1	–	1	0	1	0	0	0
r_2	1	–	0	1	0	0	0
r_3	0	0	–	0	0	1	0
r_4	1	1	0	–	0	0	0
r_5	0	0	0	0	–	0	1
r_6	0	0	1	0	0	–	0
r_7	0	0	0	0	1	0	–

Compute the Rand Index and Jaccard Coefficient by comparing pairs of distinct objects. Use the following convention:

f_{11} : same cluster and same class, f_{10} : same cluster but different class,

f_{01} : different cluster but same class, f_{00} : different cluster and different class.

The formulas are:

$$\text{RandIndex} = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}}, \qquad \text{JaccardCoefficient} = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}.$$

3. **Centroid-based cohesion and separation with normalization.**

Given are 8 two-dimensional points with coordinates shown in Table 2. The clustering result is:

$$G_1 = \{s_1, s_2\}, \qquad G_2 = \{s_3, s_4, s_5\}, \qquad G_3 = \{s_6, s_7, s_8\}.$$

Calculate the cohesion and separation for G_2 using the centroid-based formulas from the lecture slides. Since the clustering was generated by K-Means, use the cluster centroids as representatives of the clusters.

Normalize all distances needed for the calculation by

$$d_N(a, b) = \frac{d_E(a, b)}{d_{\max}}.$$

Complete the following table for the distances of each point to the centroid of its own cluster. The raw distance column is partially completed. Complete the missing raw distances and then normalize all rows.

Table 1: Distances of points from the centroid of their own cluster.

Point	Cluster	$d_E(s, \text{center}(G_i))$	$d_N(s, \text{center}(G_i))$
s_1	G_1		
s_2	G_1		
s_3	G_2	1.33	
s_4	G_2	1.05	
s_5	G_2		
s_6	G_3	1.70	
s_7	G_3	0.75	
s_8	G_3		

Finally compute the normalized centroid-to-centroid distances needed for the separation of G_2:

d_N	center(G_1)	center(G_2)	center(G_3)
center(G_1)	0		
center(G_2)		0	
center(G_3)			0

Calculate the cohesion and seperation using the formulas: The centroid-based cohesion formula

$$\text{cohesion}(G) = 1 - \frac{1}{|G|} \sum_{s \in G} d_N(s, \text{center}(G)).$$

The centroid-based separation formula is

$$\text{separation}(G) = \frac{1}{|\zeta| - 1} \sum_{G' \in \zeta,\, G' \neq G} d_N(\text{center}(G), \text{center}(G')).$$

Table 2: Point coordinates for Task 3.

ID	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8
x	1	2	5	6	8	5	7	8
y	1	3	5	6	4	1	2	1

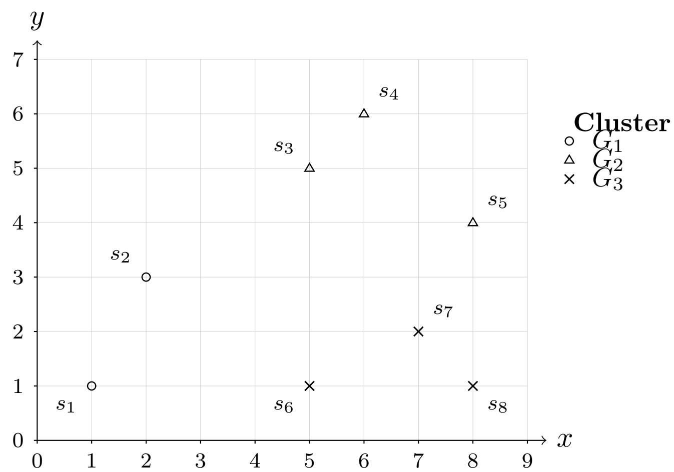

Figure 1: Clustering result for Task 3.

Exercise 9: Additional Practice Sheet – Solutions

1. **Purity with different cluster structures.**

   We use the same purity methodology as in the solution sheet. Let

$$m_{ij} = \text{number of objects in cluster } i \text{ that belong to class } j,$$

and let

$$m_i = \text{number of objects in cluster } i.$$

Then

$$p_{ij} = \frac{m_{ij}}{m_i}$$

is the probability that a member of cluster i belongs to class j.

The purity of cluster i is

$$\text{purity}(i) = \max_j p_{ij}.$$

The purity of the whole clustering is the weighted average

$$\text{purity}(Z) = \sum_{i=1}^{K} \frac{m_i}{m} \text{purity}(i),$$

where m is the total number of objects. Here,

$$m = 12.$$

For Z_A:

$$U_1 = \{a_1, a_2, b_1\}, \quad U_2 = \{a_3, b_2, b_3, b_4\}, \quad U_3 = \{a_4, c_1, c_2, c_3, c_4\}.$$

	A	B	C	m_i
U_1	2	1	0	3
U_2	1	3	0	4
U_3	1	0	4	5

For cluster U_1:

$$p_{1A} = \frac{2}{3}, \qquad p_{1B} = \frac{1}{3}, \qquad p_{1C} = \frac{0}{3} = 0.$$

Hence,

$$\text{purity}(1) = \max\left\{\frac{2}{3}, \frac{1}{3}, 0\right\} = \frac{2}{3}.$$

For cluster U_2:

$$p_{2A} = \frac{1}{4}, \qquad p_{2B} = \frac{3}{4}, \qquad p_{2C} = \frac{0}{4} = 0.$$

Hence,

$$\text{purity}(2) = \max\left\{\frac{1}{4}, \frac{3}{4}, 0\right\} = \frac{3}{4}.$$

For cluster U_3:

$$p_{3A} = \frac{1}{5}, \qquad p_{3B} = \frac{0}{5} = 0, \qquad p_{3C} = \frac{4}{5}.$$

Hence,

$$\text{purity}(3) = \max\left\{\frac{1}{5}, 0, \frac{4}{5}\right\} = \frac{4}{5}.$$

Therefore,

$$\text{purity}(Z_A) = \frac{3}{12}\cdot\frac{2}{3} + \frac{4}{12}\cdot\frac{3}{4} + \frac{5}{12}\cdot\frac{4}{5}.$$

$$\text{purity}(Z_A) = \frac{2}{12} + \frac{3}{12} + \frac{4}{12} = \frac{9}{12} = 0.75.$$

$$\boxed{\text{purity}(Z_A) = 0.75}$$

For Z_B:

$$V_1 = \{a_1, a_2\}, \quad V_2 = \{a_3, a_4, b_1\}, \quad V_3 = \{b_2, b_3\}, \quad V_4 = \{b_4, c_1, c_2\}, \quad V_5 = \{c_3, c_4\}.$$

	A	B	C	m_i
V_1	2	0	0	2
V_2	2	1	0	3
V_3	0	2	0	2
V_4	0	1	2	3
V_5	0	0	2	2

For cluster V_1:

$$p_{1A} = \frac{2}{2} = 1, \qquad p_{1B} = 0, \qquad p_{1C} = 0.$$

Hence,

$$\text{purity}(1) = 1.$$

For cluster V_2:

$$p_{2A} = \frac{2}{3}, \qquad p_{2B} = \frac{1}{3}, \qquad p_{2C} = 0.$$

Hence,

$$\text{purity}(2) = \frac{2}{3}.$$

For cluster V_3:

$$p_{3A} = 0, \qquad p_{3B} = \frac{2}{2} = 1, \qquad p_{3C} = 0.$$

Hence,

$$\text{purity}(3) = 1.$$

For cluster V_4:

$$p_{4A} = 0, \qquad p_{4B} = \frac{1}{3}, \qquad p_{4C} = \frac{2}{3}.$$

Hence,

$$\text{purity}(4) = \frac{2}{3}.$$

For cluster V_5:

$$p_{5A} = 0, \qquad p_{5B} = 0, \qquad p_{5C} = \frac{2}{2} = 1.$$

Hence,

$$\text{purity}(5) = 1.$$

Therefore,

$$\text{purity}(Z_B) = \frac{2}{12} \cdot 1 + \frac{3}{12} \cdot \frac{2}{3} + \frac{2}{12} \cdot 1 + \frac{3}{12} \cdot \frac{2}{3} + \frac{2}{12} \cdot 1.$$

$$\text{purity}(Z_B) = \frac{2}{12} + \frac{2}{12} + \frac{2}{12} + \frac{2}{12} + \frac{2}{12} = \frac{10}{12} = 0.83.$$

$$\boxed{\text{purity}(Z_B) = 0.83}$$

Conclusion:

$$\boxed{\text{purity}(Z_A) = 0.75, \qquad \text{purity}(Z_B) = 0.83.}$$

Z_B gets the higher purity. This illustrates the known problem of purity: it can favor clusterings with more, smaller clusters. In the extreme case, if each object is placed in its own singleton cluster, every cluster has purity 1.

2. **Rand Index and Jaccard Coefficient.**

The classes are

$$A = \{r_1, r_2, r_3\}, \qquad B = \{r_4, r_5\}, \qquad C = \{r_6, r_7\}.$$

The clustering is

$$H_1 = \{r_1, r_2, r_4\}, \qquad H_2 = \{r_3, r_6\}, \qquad H_3 = \{r_5, r_7\}.$$

There are 7 objects, so the total number of unordered pairs is

$$\frac{7(7-1)}{2} = 21.$$

We compare each pair according to whether it is in the same or different class and in the same or different cluster.

Pair	Same cluster?	Same class?	Category
(r_1, r_2)	1	1	f_{11}
(r_1, r_3)	0	1	f_{01}
(r_1, r_4)	1	0	f_{10}
(r_1, r_5)	0	0	f_{00}
(r_1, r_6)	0	0	f_{00}
(r_1, r_7)	0	0	f_{00}
(r_2, r_3)	0	1	f_{01}
(r_2, r_4)	1	0	f_{10}
(r_2, r_5)	0	0	f_{00}
(r_2, r_6)	0	0	f_{00}
(r_2, r_7)	0	0	f_{00}
(r_3, r_4)	0	0	f_{00}
(r_3, r_5)	0	0	f_{00}
(r_3, r_6)	1	0	f_{10}
(r_3, r_7)	0	0	f_{00}
(r_4, r_5)	0	1	f_{01}
(r_4, r_6)	0	0	f_{00}
(r_4, r_7)	0	0	f_{00}
(r_5, r_6)	0	0	f_{00}
(r_5, r_7)	1	0	f_{10}
(r_6, r_7)	0	1	f_{01}

Hence,

$$f_{11} = 1, \qquad f_{10} = 4, \qquad f_{01} = 4, \qquad f_{00} = 12.$$

Check:

$$1 + 4 + 4 + 12 = 21.$$

Rand Index:

$$\text{RandIndex} = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}} = \frac{1 + 12}{1 + 4 + 4 + 12} = \frac{13}{21} = 0.62.$$

Jaccard Coefficient:

$$\text{JaccardCoefficient} = \frac{f_{11}}{f_{11} + f_{10} + f_{01}} = \frac{1}{1+4+4} = \frac{1}{9} = 0.11.$$

$$\boxed{\text{RandIndex} = 0.62, \qquad \text{JaccardCoefficient} = 0.11.}$$

3. **Centroid-based cohesion and separation with normalization.**

The clusters are

$$G_1 = \{s_1, s_2\}, \qquad G_2 = \{s_3, s_4, s_5\}, \qquad G_3 = \{s_6, s_7, s_8\}.$$

Coordinates:

ID	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8
x	1	2	5	6	8	5	7	8
y	1	3	5	6	4	1	2	1

The normalization constant is

$$d_{\max} = d_E(s_1, s_5) = \sqrt{(1-8)^2 + (1-4)^2} = \sqrt{58}.$$

Thus,

$$d_N(a,b) = \frac{d_E(a,b)}{\sqrt{58}}.$$

Step 1: Cluster centroids.

$$\text{center}(G_1) = \left(\frac{1+2}{2}, \frac{1+3}{2}\right) = (1.50, 2.00).$$

$$\text{center}(G_2) = \left(\frac{5+6+8}{3}, \frac{5+6+4}{3}\right) = \left(\frac{19}{3}, 5\right) \approx (6.33, 5).$$

$$\text{center}(G_3) = \left(\frac{5+7+8}{3}, \frac{1+2+1}{3}\right) = \left(\frac{20}{3}, \frac{4}{3}\right) \approx (6.67, 1.33).$$

Cluster	x	y
G_1	1.50	2.00
G_2	6.33	5.00
G_3	6.67	1.33

Step 2: Distances of points from their own centroids.

Point	Cluster	$d_E(s, \text{center}(G_i))$	$d_N(s, \text{center}(G_i))$
s_1	G_1	1.12	0.15
s_2	G_1	1.12	0.15
s_3	G_2	1.33	0.18
s_4	G_2	1.05	0.14
s_5	G_2	1.94	0.26
s_6	G_3	1.70	0.22
s_7	G_3	0.75	0.10
s_8	G_3	1.37	0.18

Since the target cluster is G_2, only the rows for s_3, s_4, s_5 are needed for cohesion.

The centroid-based cohesion formula is

$$\text{cohesion}(G) = 1 - \frac{1}{|G|} \sum_{s \in G} d_N(s, \text{center}(G)).$$

Therefore,

$$\text{cohesion}(G_2) = 1 - \frac{0.18 + 0.14 + 0.26}{3}.$$
$$\text{cohesion}(G_2) = 1 - 0.19 = 0.81.$$

Exact form:

$$\text{cohesion}(G_2) = 1 - \frac{4 + \sqrt{10} + \sqrt{34}}{9\sqrt{58}}.$$

Step 3: Centroid-to-centroid distances.

$$d_E(\text{center}(G_1), \text{center}(G_2)) = \sqrt{\left(1.5 - \frac{19}{3}\right)^2 + (2-5)^2} = \frac{\sqrt{1165}}{6} \approx 5.69.$$
$$d_N(\text{center}(G_1), \text{center}(G_2)) = \frac{5.69}{\sqrt{58}} \approx 0.75.$$

$$d_E(\text{center}(G_2), \text{center}(G_3)) = \sqrt{\left(\frac{19}{3} - \frac{20}{3}\right)^2 + \left(5 - \frac{4}{3}\right)^2} = \frac{\sqrt{122}}{3} \approx 3.68.$$
$$d_N(\text{center}(G_2), \text{center}(G_3)) = \frac{3.68}{\sqrt{58}} \approx 0.48.$$

For completeness,

$$d_N(\text{center}(G_1), \text{center}(G_3)) \approx 0.68.$$

d_N	center(G_1)	center(G_2)	center(G_3)
center(G_1)	0	0.75	0.68
center(G_2)	0.75	0	0.48
center(G_3)	0.68	0.48	0

The centroid-based separation formula is

$$\text{separation}(G) = \frac{1}{|\zeta| - 1} \sum_{G' \in \zeta,\, G' \neq G} d_N(\text{center}(G), \text{center}(G')).$$

Since $\zeta = \{G_1, G_2, G_3\}$, we have $|\zeta| - 1 = 2$. Hence,

$$\text{separation}(G_2) = \frac{0.75 + 0.48}{2} = 0.62.$$

Exact form:

$$\text{separation}(G_2) = \frac{\sqrt{1165}}{12\sqrt{58}} + \frac{\sqrt{122}}{6\sqrt{58}}.$$

$$\boxed{\text{cohesion}(G_2) = 0.81, \qquad \text{separation}(G_2) = 0.62.}$$